

Digital Image Processing

Complete Exam Guide

Compiled from lecture notes

September 20, 2026

How to use this guide

Every topic from the course notes is covered below, in the order that builds understanding, with each formula typeset properly and each worked example recomputed. Passages marked **Check** are places where the original notes disagreed with the standard textbook definition; if a professor taught it differently, use that version, but be aware of the standard form for exam purposes.

Priority order for revision: (1) Intensity transforms, histogram equalization; (2) Convolution, edge detection, Canny; (3) Interpolation, padding; (4) Colour models, HSV; (5) Shape features, Hough, MLP; (6) Introduction and imaging modalities.

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1 Introduction, Pipeline, Imaging Modalities, Sensors

1.1 What is Digital Image Processing (DIP)?

A digital image is a continuous light signal converted into a pixelated matrix. DIP means processing those matrices with algorithms: mostly image-in, image-out operations (enhancement, filtering, restoration, compression).

DIP vs. Computer Vision (CV): DIP transforms an image into a better image. CV extracts meaning from an image (recognise, detect, decide).

Step	Belongs to
Original selfie	Input
Increase brightness	DIP
Remove noise	DIP
Detect face	DIP / CV boundary
Recognise person	Computer Vision
Unlock phone	Computer Vision

1.2 Image Formation Model

$$f(x, y) = i(x, y) r(x, y), \quad 0 < i(x, y) < \infty, \quad 0 < r(x, y) < 1$$

where $i(x, y)$ is illumination and $r(x, y)$ is reflectance.

Pipeline: Illumination $\rightarrow$ Reflection $f(x, y, \lambda) \rightarrow$ Optical lens $\rightarrow$ Sensor plane $f(x, y) \rightarrow$ Sampling $f(i, j) \rightarrow$ Quantization $q(i, j) \rightarrow$ Digital image $g(i, j) \rightarrow$ Encoding / Storage / Transmission.

The perforated-strip image transmission system of the 1920s is the **Bartlane cable picture transmission system** (Harry Bartholomew and Maynard McFarlane), not Frank Macfarlane Burnet (an immunologist). Confirm against the lecture slide.

1.3 Imaging Modalities

Modality	How the image forms	Key points
X-ray	Dense materials absorb more X-rays and appear white	Bones white; modern X-ray uses high-pass filtering, Digital Radiography (DR)
CT	Cross-sections reconstructed from many X-ray projections	Helical sensor approach
PET	Tracer accumulates in metabolically active tissue; gamma rays detected	Bright spots at active regions
Infrared	Wavelength $\approx 0.75 \mu\text{m}$ to $14 \mu\text{m}$	Represents temperature, not colour; fever screening, fire detection
Multispectral	Typically 4–13 spectral bands	NIR, SWIR; land-use, agriculture, disaster management

Modality	How the image forms	Key points
Hyperspectral	100–300+ narrow bands	Crop disease, mineral exploration, cancer cell detection
Microscopy	Reflective in nature	Detects crystalline deposits (stones, cholesterol)

1.4 Human Eye

- **Scotopic vision:** dim light, via rods (no colour, high sensitivity).
- **Photopic vision:** bright light, via cones (colour vision, concentrated at the fovea).
- **Glare limit:** upper brightness tolerance; subjective brightness follows a relative, log-like scale.

1.5 Sensors

CCD	CMOS
Charge shifted across chip, read at one output	Each pixel has its own amplifier and readout
Low noise, high quality	Cheaper, faster, lower power, somewhat noisier
More power, more costly	Dominant in phone cameras

1.6 Resolution and Quantization

$$L = 2^b, \quad \text{Storage (bits)} = M \times N \times b$$

- **Spatial resolution:** smallest detail the system can show.
- **Radiometric resolution:** bit depth (fine differences in light energy).
- **Image quantization:** groups continuous intensity into discrete levels; a lossy process.
- **Optical zoom** changes the lens (real detail); **digital zoom** enlarges pixels by interpolation (no new detail).

2 Interpolation, Resizing, Padding

Base example used throughout:

$$\begin{pmatrix} 10 & 20 \\ 30 & 40 \end{pmatrix}$$

2.1 Nearest Neighbour (NN) Interpolation

Copies the exact value of the closest old pixel; no new values are created. Enlarging causes a blocky / staircase effect.

$$i_{\text{old}} = i_{\text{new}} \left[\text{round} \left(\frac{r}{sf}, \frac{c}{sf} \right) \right]$$

where r, c are the new pixel's row/column and sf is the scale factor.

2×2 to 4×4 by NN:

$$\begin{pmatrix} 10 & 10 & 20 & 20 \\ 10 & 10 & 20 & 20 \\ 30 & 30 & 40 & 40 \\ 30 & 30 & 40 & 40 \end{pmatrix}$$

New cell	$r/sf, c/sf$	Rounded	Old cell	Value
(2, 2)	(1, 1)	(1, 1)	(1, 1)	10
(3, 3)	(1.5, 1.5)	(2, 2)	(2, 2)	40
(4, 4)	(2, 2)	(2, 2)	(2, 2)	40

2.2 Bilinear Interpolation

Weighted average of the 2×2 grid of nearest neighbours; linear interpolation first horizontally, then vertically. Removes the blocky effect of NN.

$$P_{\text{new}} = P_{11}(1-x)(1-y) + P_{12}(1-x)y + P_{21}x(1-y) + P_{22}xy$$

where $x, y \in [0, 1]$ are the fractional offsets.

Case $x = 0, y = 0.5$ (midpoint of top row):

$$P_{\text{new}} = 0.5P_{11} + 0.5P_{12} = \frac{P_{11} + P_{12}}{2} = \frac{10 + 20}{2} = 15$$

Centre of the grid ($x = y = 0.5$): average of all four = $\frac{10 + 20 + 30 + 40}{4} = 25$.

2×2 to 3×3 by bilinear:

$$\begin{pmatrix} 10 & 15 & 20 \\ 20 & 25 & 30 \\ 30 & 35 & 40 \end{pmatrix}$$

Worked example ($x = 0.25, y = 0.75$):

$$P = 10(0.75)(0.25) + 20(0.75)(0.75) + 30(0.25)(0.25) + 40(0.25)(0.75) = 1.875 + 11.25 + 1.875 + 7.5 = 22.5$$

2.3 Bicubic Interpolation

Blends the 16 surrounding pixels (4×4 grid) using a cubic spline curve across rows and columns. Gentler transitions, sharper details, slower than bilinear.

Bicubic is *not* inverse-distance weighting (IDW). The standard description is cubic convolution over a 4×4 neighbourhood.

Lanczos kernel (windowed sinc, a = window size):

$$L(x) = \begin{cases} \text{sinc}(x) \text{sinc}(x/a), & |x| < a \\ 0, & \text{otherwise} \end{cases}, \quad \text{sinc}(x) = \frac{\sin(\pi x)}{\pi x}$$

Method	Neighbours	Quality	Speed
Nearest neighbour	1	Blocky, jagged	Fastest
Bilinear	4 (2×2)	Smooth, slightly soft	Medium
Bicubic	16 (4×4)	Smoothest, sharpest	Slowest

2.4 Padding

Type	What is added
Zero padding (constant)	Zeros (or another constant)
Replication (clamp)	Copies of nearest edge pixel
Reflection (mirror)	Mirror image of nearby pixels
Wrap (periodic)	Pixels from the opposite side

2.5 Band Ratioing and Distance Measures

$$\text{NDVI} = \frac{\text{NIR} - \text{Red}}{\text{NIR} + \text{Red}}$$

$$D_E = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}, \quad D_4 = |x_1 - x_2| + |y_1 - y_2|, \quad D_8 = \max(|x_1 - x_2|, |y_1 - y_2|)$$

3 Intensity Transformations, Contrast, Histograms

3.1 Basic Definitions

- **Contrast:** variation in the intensity of pixels.
- **Tones:** variation in brightness/contrast/colour across an image — *highlights, midtones, shadows*.
- **Negative:** inverts intensity levels.
- **Diapositive:** identity transform (positive transparency).

$$\text{Negative (8-bit): } s = 255 - r \quad \text{General: } s = (L - 1) - r \quad \text{Diapositive: } s = r$$

3.2 Reading a Histogram

x-axis = intensity (0–255), y-axis = frequency. Mass near 0 = dark image; mass near 255 = bright image. Narrow histogram = low contrast; wide spread = high contrast. **Higher variance of the intensity distribution means higher contrast.**

3.3 Contrast Shifting vs. Stretching

Shifting: adding a constant moves all values equally (contrast unchanged, brightness changes).

Stretching: expands the intensity range.

$$B(x, y) = \frac{I_B(x, y) - I_{B, \min}}{I_{B, \max} - I_{B, \min}} \times 255$$

The multiplier for an 8-bit image is $\times 255$, not $\times 2.55$.

Example: min = 50, max = 150, pixel = 100:

$$\frac{100 - 50}{150 - 50} \times 255 = 127.5 \approx 128$$

3.4 Transformation Family

Type	Function	Effect
Linear	$g = \alpha f + \beta$	α = contrast (gain), β = brightness (bias)
Logarithmic	$g = b \log(\alpha f + 1)$	Expands dark values, compresses bright
Exponential	$g = b(e^{\alpha f} - 1)$	Expands bright values, compresses dark
Power law, $k > 1$	$g = \alpha f^k$	Darkens, expands bright range
Power law, $0 < k < 1$	$g = \alpha f^k$	Brightens, expands dark range

Textbook gamma correction: $s = cr^\gamma$.

3.5 Colour Contrast and Luminance

$$\text{Contrast} = \sqrt{(R_1 - R_2)^2 + (G_1 - G_2)^2 + (B_1 - B_2)^2}$$

$$Y = 0.29R + 0.59G + 0.11B \quad (\text{human-eye weighting: Green 59\%, Red 29\%, Blue 11\%})$$

White vs. black: $\sqrt{3 \times 255^2} \approx 441.7$.

Example (yellow, $R=255, G=255, B=0$):

$$Y = 0.29(255) + 0.59(255) + 0.11(0) = 224.4$$

3.6 Histogram Equalization

Goal: spread the histogram across the full range to increase contrast.

$$s_k = \text{round}((L - 1) \cdot \text{CDF}(r_k)), \quad \text{CDF}(r_k) = \frac{\sum_{j \leq k} n_j}{\text{total pixels}}$$

Worked example (3-bit image, $L = 8$, 42 pixels):

Original	Frequency	CDF	CDF $\times 7$	Transformed
0	0	0	0.00	0
1	4	4/42	0.67	1
2	5	9/42	1.50	2
3	9	18/42	3.00	3
4	9	27/42	4.50	5
5	12	39/42	6.50	7
6	3	42/42	7.00	7
7	0	42/42	7.00	7

Frequencies at levels 3 and 4 must be 9 and 9 (not 8 and 10), since they must sum with the earlier counts to give CDF numerators 18/42 and 27/42: $4 + 5 + 9 = 18$, then $18 + 9 = 27$.

Result: levels 5, 6, 7 merge into output level 7 (with $12 + 3 + 0 = 15$ pixels); equalization cannot make a perfectly flat histogram in a discrete image. New histogram: level 1: 4, level 2: 5, level 3: 9, level 5: 9, level 7: 15 (total 42).

4 Spatial Filtering, Kernels, Edge Detection

4.1 Convolution

$$g(x, y) = \sum_{i=-m}^{+m} \sum_{j=-n}^{+n} K(i, j) f(x - i, y - j)$$

For a 3×3 kernel, $m = n = 1$.

Mean filter example: 3×3 window, all pixels 30 except the centre (0), kernel = all-ones /9:

$$\frac{30 \cdot 1 \times 8 + 0 \cdot 1}{9} = \frac{240}{9} \approx 26.67$$

Filter type	Promotes	Kernel sum
Low-pass (mean, Gaussian)	Smooth regions, removes noise	1 (after normalising)
High-pass (Prewitt, Sobel, Laplacian, LoG)	Edges, detail	0 (zero-sum)

4.2 Prewitt Kernel

Vertical-edge kernel:

$$K_v = \begin{pmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{pmatrix} \quad K_h = \begin{pmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix}$$

Example (top rows 200, bottom rows 50, horizontal kernel):

$$-(200 \times 3) + (50 \times 3) = -600 + 150 = -450$$

4.3 Sobel Kernel

Prewitt with extra weight (2) on the centre row/column:

$$K_v = \begin{pmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{pmatrix} \quad K_h = \begin{pmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{pmatrix}$$

Example (top 200, bottom 50):

$$-(200 \times 4) + (50 \times 4) = -800 + 200 = -600$$

Diagonal edge kernels:

$$\begin{pmatrix} 0 & -1 & -1 \\ 1 & 0 & -1 \\ 1 & 1 & 0 \end{pmatrix} \quad \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & -1 \\ 0 & -1 & -1 \end{pmatrix}$$

4.4 Gaussian Kernel

$$\frac{1}{16} \begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{pmatrix}$$

$$G(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

4.5 Laplacian Filter

4-neighbour version:

$$\begin{pmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{pmatrix}$$

8-neighbour version:

$$\begin{pmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{pmatrix}$$

The centre of the 8-neighbour kernel must be **8**, not 0, so the kernel sums to zero (a required property of a high-pass filter).

Sharpening: if centre is -4 , use $g = f - \nabla^2 f$; if centre is $+4$, use $g = f + \nabla^2 f$. High-boost: $g = f + k(f - \bar{f})$, $k > 1$.

4.6 Laplacian of Gaussian (LoG)

$$\begin{pmatrix} 0 & 0 & -1 & 0 & 0 \\ 0 & -1 & -2 & -1 & 0 \\ -1 & -2 & 16 & -2 & -1 \\ 0 & -1 & -2 & -1 & 0 \\ 0 & 0 & -1 & 0 & 0 \end{pmatrix}$$

Sum of all entries = 0, so it is a high-pass filter (Gaussian alone is all-positive, low-pass).

4.7 Edge Strength, Direction, and Canny

$$\text{Edge strength} = \sqrt{\left(\frac{\partial I}{\partial x}\right)^2 + \left(\frac{\partial I}{\partial y}\right)^2}, \quad \theta = \tan^{-1}\left(\frac{dy}{dx}\right)$$

Canny steps:

1. Smooth with a Gaussian filter.
2. Compute gradient strength and direction (Sobel).
3. Non-maximum suppression (thin edges to local maxima along the gradient direction).
4. Double threshold and **hysteresis** (connect edges).

Non-max suppression example:

$$\begin{pmatrix} 270 & 40 & 85 \\ 119 & 450 & 60 \\ 90 & 220 & 100 \end{pmatrix} \longrightarrow \begin{pmatrix} 0 & 0 & 0 \\ 0 & 450 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

The correct term is **hysteresis**, not “hypothesis.”

Hysteresis rule: gradient above high threshold = strong edge (keep); below low threshold = discard; in between = keep only if connected to a strong edge.

4.8 CNNs: Padding and Stride

$$\text{Output size} = \left\lfloor \frac{N + 2p - k}{s} \right\rfloor + 1$$

where N = input size, k = kernel size, p = padding, s = stride.

Example: $N = 256, k = 3, p = 0, s = 1$: $(256 - 3)/1 + 1 = 254$.

Example: $N = 224, k = 3, p = 1, s = 2$: $\lfloor (224 + 2 - 3)/2 \rfloor + 1 = \lfloor 111.5 \rfloor + 1 = 112$.

5 Colour: False Colour Composites and HSV

5.1 False Colour Composite (FCC)

A colour image built by assigning spectral bands (including wavelengths beyond the visible spectrum, e.g. NIR) to the R, G, B display channels.

$$\# \text{composites} = 3! = 6, \quad \# \text{true colour} = 1, \quad \# \text{false colour} = 3! - 1 = 5$$

Standard remote-sensing FCC: NIR→R, Red→G, Green→B, so healthy vegetation appears bright red (vegetation reflects IR strongly).

Why HSV is used: separately equalizing R, G, B distorts colour balance. Converting to HSV and equalizing only V (brightness) keeps hue and saturation intact.

5.2 HSV Model

- **Hue (H):** angle on the colour wheel, 0° to 360° .
- **Saturation (S):** 0 (grayish) to 1 (pure colour).
- **Value (V):** brightness.

Hue wheel: Red 0° , Yellow 60° , Green 120° , Cyan 180° , Blue 240° , Magenta 300° .

Yellow is at 60° , not 90° (the six primaries/secondaries are 60° apart).

5.3 RGB to HSV Conversion

$$R' = \frac{R}{255}, \quad G' = \frac{G}{255}, \quad B' = \frac{B}{255}$$

$$C_{\max} = \max(R', G', B'), \quad C_{\min} = \min(R', G', B'), \quad \Delta = C_{\max} - C_{\min}$$

$$V = C_{\max}$$

$$S = \begin{cases} 0, & C_{\max} = 0 \\ \frac{\Delta}{C_{\max}}, & \text{otherwise} \end{cases}$$

$$H = \begin{cases} 0^\circ, & \Delta = 0 \\ 60^\circ \times \left(\frac{G' - B'}{\Delta} \bmod 6 \right), & C_{\max} = R' \\ 60^\circ \times \left(\frac{B' - R'}{\Delta} + 2 \right), & C_{\max} = G' \\ 60^\circ \times \left(\frac{R' - G'}{\Delta} + 4 \right), & C_{\max} = B' \end{cases}$$

Worked example 1 (pink): given $H = 0^\circ, S = 0.5, V = 1$. $V = C_{\max} = 1 \Rightarrow R' = 1$ ($R = 255$). $S = 1 - C_{\min}/C_{\max} = 0.5 \Rightarrow C_{\min} = 0.5$ ($G' = B' = 0.5$).

$$(R, G, B) = (255, 128, 128) \approx \text{pink}$$

Pink is (255, 128, 128), not (255, 255, 255) (that is white).

Worked example 2: RGB (200, 100, 50) $\rightarrow$ HSV.

$$R' = 0.784, \quad G' = 0.392, \quad B' = 0.196, \quad C_{\max} = 0.784, \quad C_{\min} = 0.196, \quad \Delta = 0.588$$

$$V = 0.784, \quad S = \frac{0.588}{0.784} = 0.75$$

$$H = 60^\circ \times \left(\frac{0.392 - 0.196}{0.588} \bmod 6 \right) = 60^\circ \times 0.333 = 20^\circ$$

Result: $H = 20^\circ, S = 0.75, V = 0.784$ (orange-brown).

6 Shape Features, Boundary Tracing, Hough Transform, MLP

6.1 Natural vs. Man-made Objects

Natural	Man-made
More corners/edges, relatively	Fewer corners/edges
Rarely perfectly symmetric	Highly symmetric
Irregular, complex	Regular, geometric
Fractal in nature	Scale-dependent

6.2 Shape Descriptors

$$\text{Circularity} = \frac{4\pi A}{P^2}$$

A =area, P =perimeter. Circularity = 1: perfect circle. Square = $\pi/4 \approx 0.785$; equilateral triangle ≈ 0.605 .

$$\kappa = \frac{d\theta}{ds}, \quad R = \frac{1}{\kappa}$$

$$\text{Edge density} = \frac{N}{A}$$

κ = curvature (rate of turning per unit path length); straight line $\kappa = 0$.

Convex hull: smallest enclosing boundary with no indentations. **Solidity** = object area / convex hull area.

6.3 Moore Neighbour Tracing

Traces the boundary of a binary object **clockwise** using the 8 surrounding (Moore) neighbours. Produces a boundary coordinate list.

Procedure: (1) find first object pixel, (2) note the backtrack pixel, (3) examine 8 neighbours clockwise from backtrack, (4) move to first object pixel found, update backtrack, (5) stop at start pixel.

Shape	Boundaries	Why
Hollow “O”	2	Outer + inner (hole)
Hollow “8”	3	Outer + two holes
“8” one loop filled	2	Outer + one hole

6.4 Angle Method (Natural vs. Man-made)

$$\theta_i = \tan^{-1}\left(\frac{\Delta y}{\Delta x}\right), \quad \frac{d\theta}{ds}$$

Natural: $d\theta$ constantly changing. Man-made: $d\theta \approx 0$ along straight paths with sudden spikes at corners (often near 90° shifts).

6.5 Hough Transform

Isolates straight lines and circles in a binary edge map using a global voting procedure in parameter space; robust to broken/noisy edges.

$$\rho = x \cos \theta + y \sin \theta$$

ρ = perpendicular distance from origin, θ = angle of that perpendicular.

Algorithm: (1) each point (x, y) maps to a sinusoid in (ρ, θ) space; (2) accumulator array counts votes; (3) each edge pixel votes for all (ρ, θ) lines through it; (4) peaks in the accumulator give detected lines.

6.6 MLP Classifier

$$y = f\left(\sum_i x_i w_i + b\right)$$

x_i =inputs, w_i =weights, b =bias, f =activation (ReLU, sigmoid).

Input features: number of straight lines (Hough); variance of $d\theta$ (Moore tracing); peak energy at $\theta + 90^\circ$; texture entropy (GLCM).

$$\text{Entropy} = - \sum_i \sum_j p(i, j) \log_2 p(i, j)$$

7 Formula Sheet

Topic	Formula
Levels and storage	$L = 2^b$; storage = $M \times N \times b$ bits
Image model	$f(x, y) = i(x, y) r(x, y)$
Negative	$s = (L - 1) - r$; 8-bit: $s = 255 - r$
Contrast stretch	$\frac{I - I_{\min}}{I_{\max} - I_{\min}} \times 255$
Log/exp/power	$g = b \log(\alpha f + 1)$; $g = b(e^{\alpha f} - 1)$; $g = \alpha f^k$
Equalization	$s_k = \text{round}((L - 1) \cdot \text{CDF}(r_k))$
Convolution	$g(x, y) = \sum \sum K(i, j) f(x - i, y - j)$
NN interpolation	$i_{\text{old}} = i_{\text{new}}[\text{round}(r/sf, c/sf)]$
Bilinear	$P = P_{11}(1-x)(1-y) + P_{12}(1-x)y + P_{21}x(1-y) + P_{22}xy$
CNN output size	$\left\lfloor \frac{N + 2p - k}{s} \right\rfloor + 1$
Edge strength/direction	$\sqrt{G_x^2 + G_y^2}$; $\theta = \tan^{-1}(G_y/G_x)$
Gaussian kernel 3×3	$\frac{1}{16} \begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{pmatrix}$
Laplacian (4-neighbour)	$\begin{pmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{pmatrix}$
HSV	$V = C_{\max}$; $S = \Delta/C_{\max}$; $H = 60^\circ \times (\dots)$
Circularity	$4\pi A/P^2$
Curvature	$\kappa = d\theta/ds$; $R = 1/\kappa$
Edge density	N/A
Hough line	$\rho = x \cos \theta + y \sin \theta$
MLP neuron	$y = f(\sum x_i w_i + b)$
NDVI	$(\text{NIR} - \text{Red})/(\text{NIR} + \text{Red})$
Distances	$D_E = \sqrt{\Delta x^2 + \Delta y^2}$; $D_4 = \Delta x + \Delta y $; $D_8 = \max(\Delta x , \Delta y)$

8 Practice Questions

1. Negative of pixel 87 (8-bit): $255 - 87 = \mathbf{168}$.
2. Bilinear at $x=0.25, y=0.75$ for $\begin{pmatrix} 10 & 20 \\ 30 & 40 \end{pmatrix}$: **22.5**.
3. Contrast-stretch pixel 120, min 60, max 180: $\frac{120 - 60}{180 - 60} \times 255 = \mathbf{127.5}$.
4. 256×256 image, 5×5 kernel, stride 1, no padding: $256 - 5 + 1 = \mathbf{252 \times 252}$.
5. 224×224 image, 3×3 kernel, padding 1, stride 2: $\lfloor (224 + 2 - 3)/2 \rfloor + 1 = \mathbf{112 \times 112}$.
6. Sobel horizontal response, 100 above 20: $-(100 \times 4) + (20 \times 4) = \mathbf{-320}$.
7. HSV of pure green (0, 255, 0): $V = 1, S = 1, H = 60 \times (0 + 2) = \mathbf{120^\circ}$.
8. Hough ρ for point (3, 4) at $\theta = 0^\circ, 90^\circ, 45^\circ$: **3, 4, 4.95**.
9. Circularity of a square, side 4: $A = 16, P = 16$, so $4\pi(16)/256 = \pi/4 \approx \mathbf{0.785}$.
10. Storage, 1024×768 image, 8-bit gray vs. 24-bit colour: **768 KB** and **2.25 MB**.
11. Mean filter centre of $\begin{pmatrix} 10 & 20 & 30 \\ 40 & 50 & 60 \\ 70 & 80 & 90 \end{pmatrix}$: sum = 450, $450/9 = \mathbf{50}$.
12. Number of false colour composites from 3 bands: $3! - 1 = \mathbf{5}$.